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(ICRTB 2024)

January 19-20, 2024

ABSTRACTS

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OP-013

Insights into *trichogramma chilonis* olfactory signalling: molecular cloning and expression profiling of pheromone binding proteins.

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Abstract: Chemical pesticides risk environmental pollution, harm to non-target species, and development of resistance in pests. In contrast, biological pesticides, like *Trichogramma*, offer targeted, sustainable solutions with minimal ecological impact and reduced potential for resistance. *Trichogramma chilonis*, a widely utilized biocontrol agent in India, was the focus of our study. Using the Illumina pair-ended sequencing platform, we conducted the first RNA-Sequence analysis of adult *T. chilonis*. The sequencing generated 18,372,639 high-quality reads, which were de novo assembled into 24,488 transcripts. *T. chilonis* specimens were mass reared on *Corcyra cephalonica* eggs, and total RNA isolation and mRNA purification were carried out. A specific gene, the Pheromone Binding Protein gene, was amplified using gene-specific primers, cloned into a TA vector, and transformed into DH5α cells. Positive clones were identified through colony PCR, and the inserted gene was confirmed by sequencing after restriction release. The obtained sequence underwent bioinformatic characterization, and the full-length gene was codon-optimized. The synthetic, codon-optimized gene was then cloned into a pET28a sumo vector and transformed into BL21 *E. coli* cells. Positive clones were confirmed by colony PCR. The expressed recombinant protein was purified and subjected to analysis using SDS PAGE. Confirmation of protein expression was further validated through western blotting. This comprehensive study provides valuable insights into the molecular aspects of *Trichogramma chilonis* and lays the groundwork for potential applications in pest management and crop protection.

Keywords: *Trichogramma*, cloning, biocontrol, pheromone binding protein.

ISBN: 978-93-6039-051-8

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ISBN 978-93-6039-051-8